# Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

# Materials

| Antibodies | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| For commercial reagents, provide supplier name, catalogue number and RRID, if available | No antibodes detected.<br>Please add identifiers for all resources where possible | |

| Cell Materials | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| **Cell lines:** Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID | No cell lines detected<br>Please add identifiers for all resources where possible | |
| **Primary cultures:** Provide species, strain, sex of origin, genetic modification status. | Not currently checked by SciScore | |

| Experimental Animals | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| **Laboratory animals:** Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID | No organisms detected<br>Please add identifiers for all resources where possible | |
| **Animal observed in or captured from the field:** Provide species, sex and age where possible | Not currently checked by SciScore | |
| **Model organisms:** Provide Accession number in repository (where relevant) OR RRID | See laboratory animals section for information. | |

| Plants and microbes | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| **Plants:** provide species and strain, unique accession number if available, and source (including location for collected wild specimens) | Not currently checked by SciScore | |
| **Microbes:** provide species and strain, unique accession number if available, and source | Not currently checked by SciScore | |

| Human research participants | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval. | Trials 843 (ClinicalTrials.gov: NCT02577003) and 849 (ClinicalTrials.gov: NCT02564211) were conducted at 25 and 12 trial centres (Supplemental Appendix) in Japan between November 2015 and November 2016, and between October 2015 and March 2017, respectively, in accordance with the principles of Good Clinical Practice and were approved by the appropriate institutional review boards and regulatory agencies. | |
| Provide statement confirming informed consent obtained from study participants. | Written informed consent was obtained from all study patients. | |
| Report on age and sex for all study participants. | **Age:**For both trials, at screening, eligible patients were male or female, aged **Age:**≥**Age:**20 years, with T2D being treated with diet and exercise therapy and meeting either of the following additional criteria: on a stable dose of sitagliptin 50 mg once daily and any single additional OHA or low-dose dual combination therapy prior to screening, with HbA1c **Age:**≥**Age:**6.5% and **Age:**≤**Age:**9.0% (Group A, which expands the sitagliptin monotherapy population after discontinuation of the additional OHA[s] | |

| Human research participants | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| | and the subsequent medication stabilization period); or on a stable dose of sitagliptin 50 mg once daily and not on any additional OHAs prior to screening, with HbA1c **Age:**≥**Age:**7.0% and **Age:**≤**Age:**10.0%, and, for Trial 849, FPG **Age:**≤**Age:**230 mg/dL (Group B).<br>**Sex:**For both trials, at screening, eligible patients were male or female, aged **Sex:**≥**Sex:**20 years, with T2D being treated with diet and exercise therapy and meeting either of the following additional criteria: on a stable dose of sitagliptin 50 mg once daily and any single additional OHA or low-dose dual combination therapy prior to screening, with HbA1c **Sex:**≥**Sex:**6.5% and **Sex:**≤**Sex:**9.0% (Group A, which expands the sitagliptin monotherapy population after discontinuation of the additional OHA[s] and the subsequent medication stabilization period); or on a stable dose of sitagliptin 50 mg once daily and not on any additional OHAs prior to screening, with HbA1c **Sex:**≥**Sex:**7.0% and **Sex:**≤**Sex:**10.0%, and, for Trial 849, FPG **Sex:**≤**Sex:**230 mg/dL (Group B). | |

# Design

| Study protocol | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| For clinical trials, provide the trial registration number OR cite DOI in manuscript. | NCT02577003<br>NCT02564211 | |

| Laboratory protocol | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| Provide DOI or other citation details if detailed step-by-step protocols are available. | Not detected. | |

| Experimental study design (statistics details) | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| State whether and how the following have been done, or if they were not carried out | | |
| Sample size determination | With a sample size of 69 patients per arm, Trial 843 was estimated to have 90% power to detect a true treatment difference of 0.5% in change from baseline in HbA1c at Week 24, assuming a true standard deviation (SD) of 0.85% ($\alpha$ = 0.05, two-sided) and a discontinuation rate of 10%. | |
| Randomization | 2 METHODS 2.1 Study designs Trial 843 was randomized, placebo-controlled and double-blind (Supplemental Figure S2A). | |
| Blinding | not detected. | |
| inclusion/exclusion criteria | Patients were excluded from the trials if they had type 1 diabetes or a history of ketoacidosis, unstable diabetic retinopathy, poorly controlled hypertension (systolic blood pressure ≥160 mm Hg or diastolic blood pressure ≥100 mm Hg), significant cardiovascular disease, active liver disease, renal disease or urological disorders, a history of malignancy or haematological disorders; if they had been treated with insulin or thiazolidinediones within 12 weeks prior to screening or with SGLT2 inhibitors at any time; or if they required treatment with systemic steroids. | |

| Sample definition and in-laboratory replication | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| State number of times the experiment was replicated in laboratory | Not detected. | |
| Define whether data describe technical or biological replicates | Not detected. | |

| Ethics | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval. | Trials 843 (ClinicalTrials.gov: NCT02577003) and 849 (ClinicalTrials.gov: NCT02564211) were conducted at 25 and 12 trial centres (Supplemental Appendix) in Japan between November 2015 and November 2016, and between October 2015 and March 2017, respectively, in accordance with the principles of Good Clinical Practice and were approved by the appropriate institutional review boards and regulatory agencies. | |
| Studies involving experimental animals: State details of authority granting ethics approval (IRB or | Not detected. | |

| Ethics | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| equivalent committee(s), provide reference number for approval. | | |
| Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why. | Not detected. | |

| Dual Use Research of Concern (DURC) | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval | Not currently checked by SciScore | |

## Analysis

| Attrition | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance. | not detected. | |

| Statistics | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| Describe statistical tests used and justify choice of tests. | In Trial 849, the number and percentage of patients achieving HbA1c <7.0% at various times during the study, and the corresponding 95% CI (based on the method of Clopper and Pearson 17 ), were calculated. | |

| Data availability | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| State whether newly created datasets are available, including protocols for access or restriction on access. | Not detected. | |
| If data are publicly available, provide accession number in repository or DOI or URL. | Not detected. | |
| If publicly available data are reused, provide accession number in repository or DOI or URL, where possible. | Not detected. | |

| Code availability | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| For all newly generated code and software essential for replicating the main findings of the study: | | |
| State whether the code or software is available. | Not detected. | |
| If code is publicly available, provide accession number in repository, or DOI or URL. | Not detected. | |

## Analysis

| Adherence to community standards | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR. | | |
| State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript. | Not currently checked by SciScore | |